

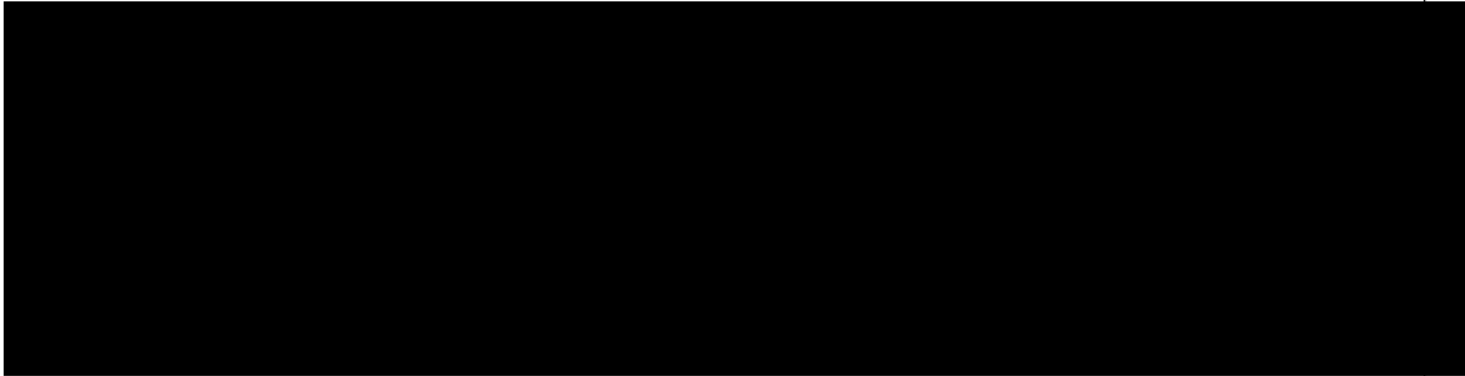
PUBLIC
EXHIBIT 54

U.S. Patent No. 10,062,748 Domestic Industry Chart

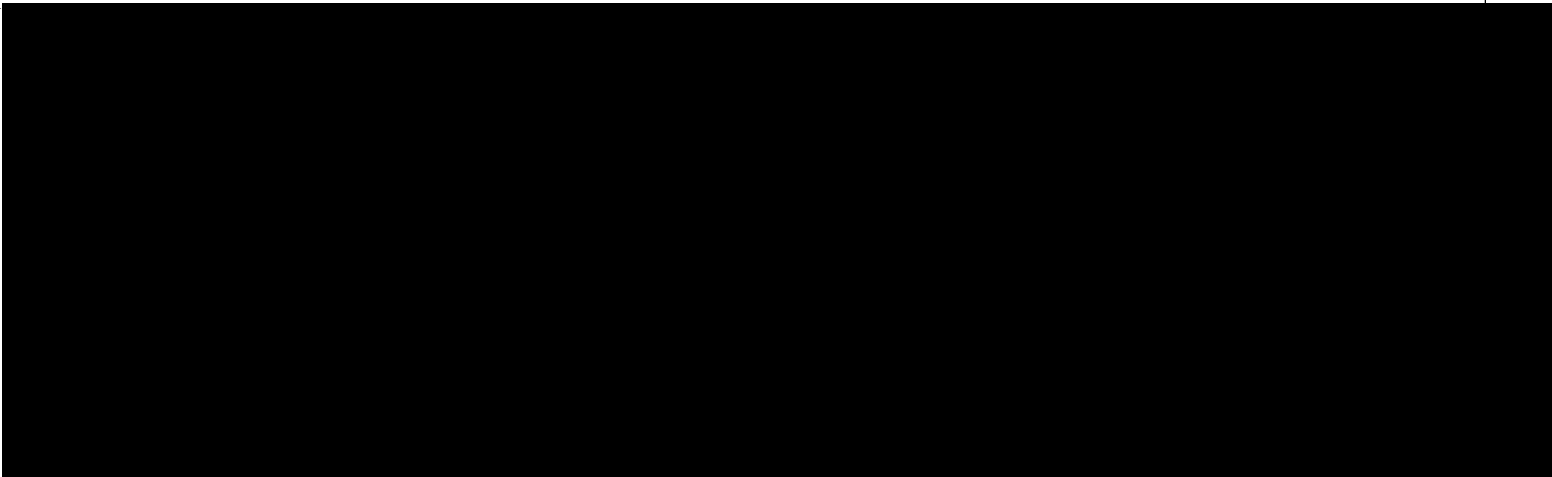
Title: Segmented guard-ring and chip edge seals

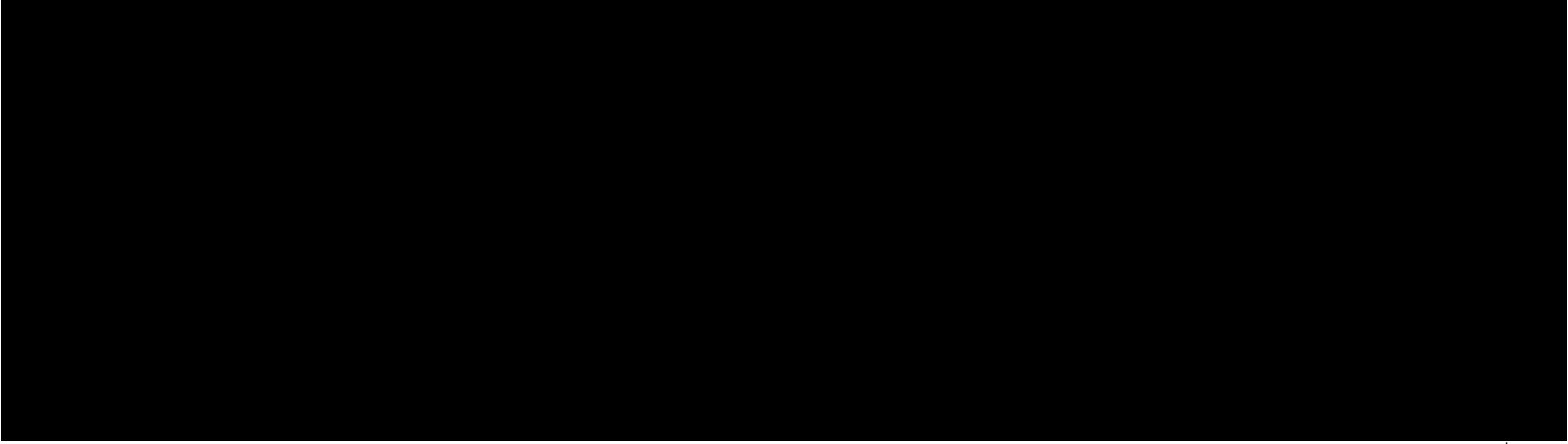
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Abstract: The present disclosure relates to semiconductor structures and, more particularly, to segmented guard-ring and chip edge seals and methods of manufacture. The structure includes: a guard ring structure formed in a low-k dielectric material; and an edge seal structure formed through the low-k dielectric material to at least a substrate underneath the low-k dielectric material.

Claim	'748 Domestic Industry Products
[1pre] A structure comprising:	The '748 Domestic Industry Products include products manufactured using GlobalFoundries' CSOI8SW process.¹ To the extent the preamble is limiting, the '748 Domestic Industry Products comprise a structure, as explained further in the following limitations. For example, the reverse engineering analysis below shows a cross section of a device manufactured using GlobalFoundries' 8SW process that contains a structure, as explained in the limitations below. 
[1a] a guard ring structure formed in a	The '748 Domestic Industry Products include a guard ring structure that is formed in a material that has a low dielectric constant.

¹ GlobalFoundries reserves the right to identify different or additional '748 Domestic Industry Products and amend or supplement these charts after further investigation and discovery, claim construction, or other meriting circumstances.

low-k dielectric material; and	For example, the cross section below (corresponding to the red box in the reverse engineering analysis above) s 
[1b] an edge seal structure formed through the low-k dielectric material to at least a substrate underneath the low-k dielectric material,	The '748 Domestic Industry Products comprise an edge seal structure formed through the low-k dielectric material to at least a substrate underneath the low-k dielectric material. For example, the cross section below shows an edge seal structure (indicated by red arrow) that is formed through th 

[1c] wherein the substrate is a silicon on insulator substrate and the edge seal structure terminates below the low-k dielectric material.	The '748 Domestic Industry Products comprise a silicon on insulator substrate and the edge seal structure terminates below the low-k dielectric material. For example, the cross section below shows an edge seal structure that terminates below the low-k dielectric material (illustrated by the red arrow). The reverse engineering analysis shows that the substrate is a silicon on insulator substrate. 
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